A Comparative Study of Generative Adversarial Networks Across Synthetic, Medical, Cybersecurity, and Creative Data Domains

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GitHub link: https://github.com/thulluri-ramya/Generative-modelling-case-study-

1. Introduction:

In generative modelling, the objective is to generate new data samples that are comparable to a dataset that has already been provided. GANs, which stand for Generative Adversarial Networks, are among the most widely used approaches for this purpose. Two neural networks are used to construct them: a discriminator, which is responsible for attempting to differentiate between created samples and actual samples, and a generator, which is responsible for producing artificial data. Together, these two components are trained through a competitive process that enables the model to gradually improve its capacity to deliver outcomes that are more realistic.

In this research, GANs are utilized and evaluated on a variety of different types of data. During the first part of the process, a GAN is trained on fundamental two-dimensional distributions in order to gain an understanding of how the model acquires fundamental patterns. In Part 2, the method is expanded to include practical applications, such as the generation of medical images by utilizing the BloodMNIST dataset, the generation of synthetic network traffic for the purpose of cybersecurity, and the generation of sketches by utilizing the QuickDraw pizza dataset.

In this work, the behavior of GANs is demonstrated across a variety of domains and classifications of data. It is possible that their performance will change when applied to datasets that are more intricate or noisy, yet they function effectively when applied to organized picture data. In order to improve model design and successfully implement generative models in real-world scenarios, it is essential to have a solid understanding of these distinctions.

2. Sine Wave GAN:

A Generative Adversarial Network was utilized on a simple sine-wave distribution to understand how GANs learn essential data patterns. The training data consisted of two-dimensional points sampled from a sine function, with x-values generated randomly and corresponding y-values conforming to the sine curve. This produced a smooth, uninterrupted distribution suitable for evaluating the model's capacity to discern an underlying mathematical framework. The generator was designed to accept random noise as input and generate synthetic data points, while the discriminator was meant to distinguish between genuine and generated samples. Both networks were simultaneously trained using binary cross-entropy loss and the Adam optimizer.

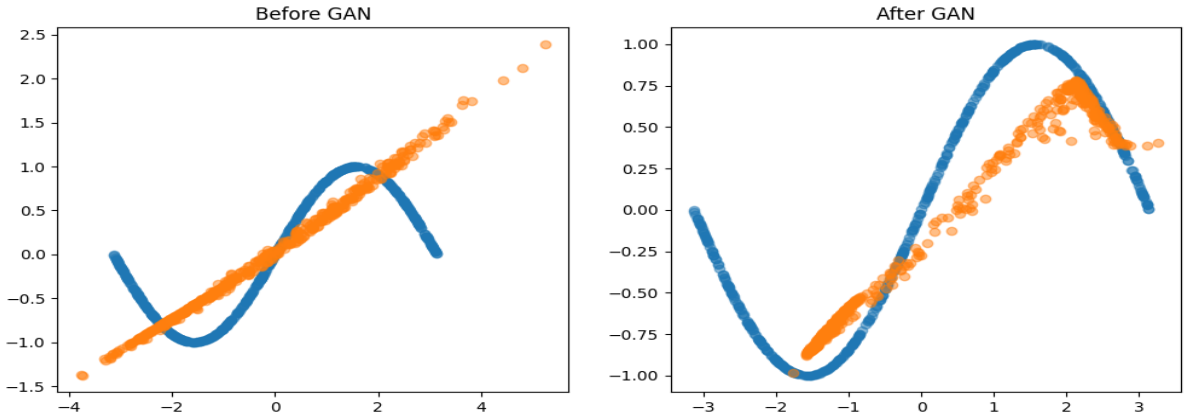

Figure 1: Real vs generated samples for the sine-wave distribution before and after GAN training.

Figure 1 demonstrates that the GAN effectively captured the overall contour of the sine-wave distribution post-training. Before training, the generated samples failed to adhere to the sine pattern and appeared as a simple linear distribution. Consequently, after training, the generated points begin to align with the sine curve, indicating that the generator has internalized key characteristics of the input. Despite the fit being imperfect and various areas deviating off the curve, the primary structure is clearly delineated. This signifies that the model comprehensively understood the underlying distribution to an acceptable extent.

3. Gaussian Mixture GAN:

A sophisticated two-dimensional distribution was represented by a mixture of Gaussian clusters to evaluate the efficacy of the GAN on non-linear and multi-modal data. Unlike the sine-wave dataset, this distribution consists of multiple discrete clusters, hindering the generator's capacity to cover all regions of the data space. The dataset consisted of many Gaussian components located at different positions, requiring the model to learn the location and spread of each cluster.

Improvements were made to the GAN architecture and training techniques to enhance performance. These included alterations to the network architecture and training parameters to better effectively capture the complexities of the data. The model was trained using binary cross-entropy loss and the Adam optimizer, while the training process was monitored using the loss curves of the generator and discriminator.

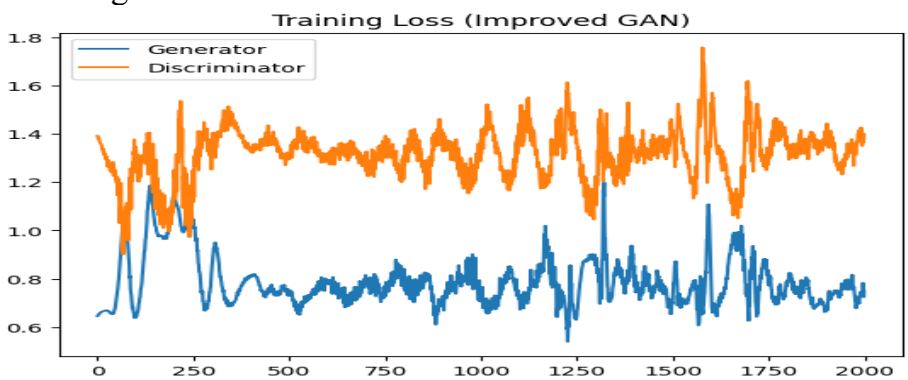

Figure 2: Training loss of generator and discriminator

Figure 2 demonstrates that the training losses display variability during the process, a typical trait of GANs due to the adversarial nature of their training. The generator loss is rather stable but shows occasional spikes, while the discriminator loss varies, indicating that both networks are continuously adapting to each other. This signifies that the model did not attain full convergence but maintained a dynamic equilibrium throughout training.

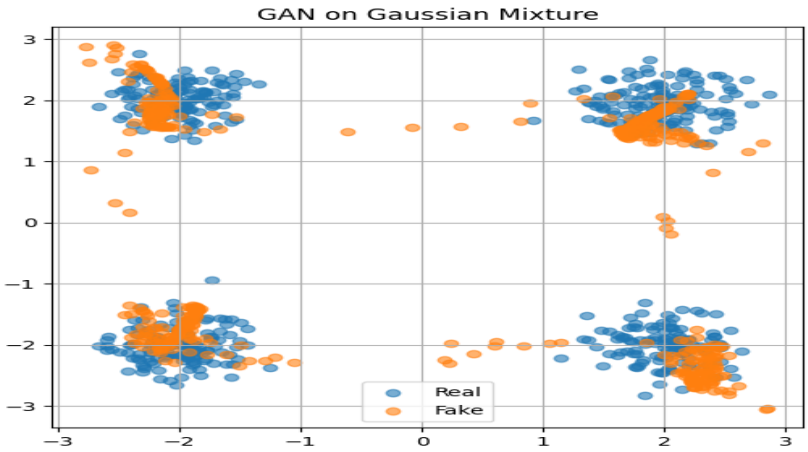

Figure 3: Real vs generated samples for Gaussian mixture

The samples generated, depicted in **Figure 3**, demonstrate that the GAN effectively attained a partial comprehension of the Gaussian mixture's configuration. The model successfully discerned multiple clusters and generated points in the relevant general regions. However, the generated data does not precisely reflect the true distribution, and certain areas appear to be scattered throughout clusters or unevenly concentrated. This indicates that while the generator understood the overall structure, it struggled to precisely represent the specific distribution of each Gaussian component.

This experiment illustrates that GANs have more significant difficulties in simulating multi-modal distributions compared to simple continuous functions. While the model demonstrates some efficacy in identifying cluster locations, it inadequately captures the density and boundaries of the data. Possible improvements may include increasing model capacity, prolonging training duration, or utilizing more advanced GAN variants to better handle complex distributions.

4. BloodMNIST (Medical GAN):

For the purpose of producing synthetic medical images of blood cells, a Generative Adversarial Network was utilized on the BloodMNIST dataset. It is possible to use the dataset for image-based generative modeling because it is composed of small RGB images that illustrate many types of blood cells. Based on the evidence shown in **Figure 4**, it can be seen that the collection possesses distinct spatial characteristics and ordered visual patterns. Because of its spatial characteristics, a Deep Convolutional GAN (DCGAN) architecture was chosen. This is because convolutional layers are able to successfully collect visual

information and textures that are local to the image.

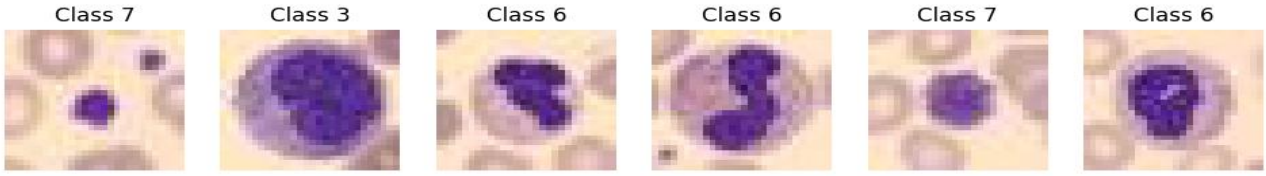

Figure 4: Exemplar photos from the BloodMNIST dataset

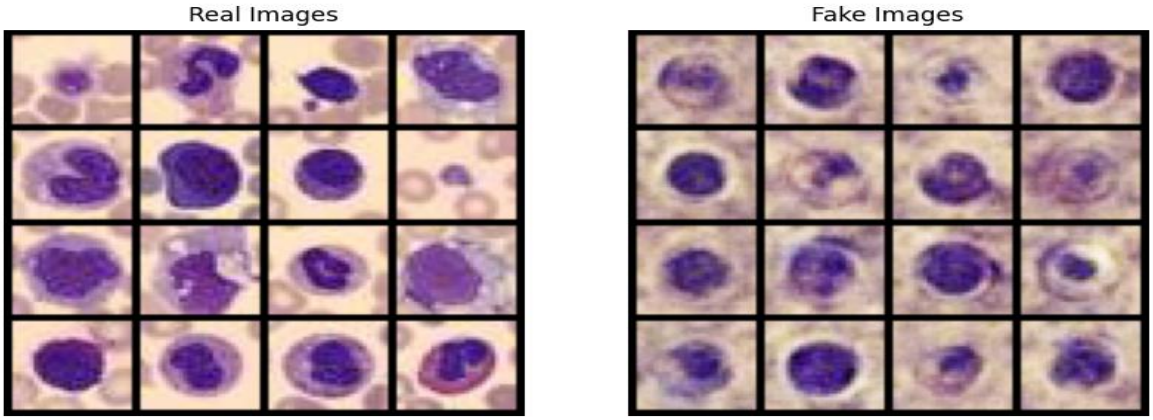

Figure 5: Blood cell pictures produced by GAN

The generator network was developed with the purpose of converting random noise vectors into image samples, whilst the discriminator network was intended to determine whether or not given images were authentic or fake. The Adam optimizer and binary cross-entropy loss were both utilized throughout the training process of both networks simultaneously. As shown in **Figure 6**, loss curves were used to monitor the training process. These curves were used for both the generator and the discriminator.

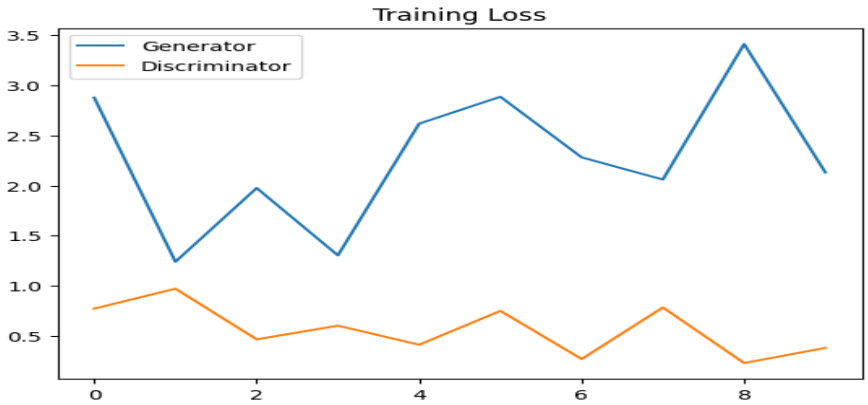

Figure 6: Curves of training loss for the generator and discriminator

As a result of the adversarial interactions that occur between the two networks, the loss curves exhibit variations over time, which is a trait that is typical of GAN training. Clearly, the GAN was successful in capturing the important characteristics of the blood cell images, as seen by the images that were generated and displayed in **Figure 5**. There are some examples that have been generated that have similarities to real cells in terms of their overall form and the distribution of colors; nevertheless, many of the images appear to be fairly blurry and lack fine detail. With this information, it appears that the model was able to correctly discern overarching patterns, but it had difficulty producing structures of a high grade.

The findings indicate that GANs are useful in providing medical picture data; nevertheless, the quality of the images is dependent on the stability of the training and the proficiency of the model. The realism of the samples that are created could be improved through the implementation of improvements such as expanding the training epochs, constructing deeper architectures, or employing conditional Gaussian neural networks. The purpose of this study is to illustrate the possible application of GANs for data augmentation in medical situations, where the generation of additional training data could be for the better.

5. Cybersecurity:

A Generative Adversarial Network was employed on the CICIDS 2017 dataset to generate synthetic network traffic data for cybersecurity assessment. This dataset consists of high-dimensional tabular data representing network properties, unlike previous tasks that included images. The dataset includes both innocuous traffic and attack patterns, such as distributed denial-of-service (DDoS) activity. Due to the absence of spatial organization in tabular data, a fully linked GAN architecture was utilized instead of a convolutional model.

Figure 7: Training loss trajectories for the generator and discriminator (CICIDS dataset)

The generator was designed to produce synthetic feature vectors from random noise, while the discriminator aimed to distinguish between genuine and fabricated network traffic samples. Both models were trained using binary cross-entropy loss and the Adam optimizer. The training process was seen through loss curves, depicted in **Figure 7**, which display fluctuations typical of GAN training due to the adversarial interaction between the generator and discriminator.

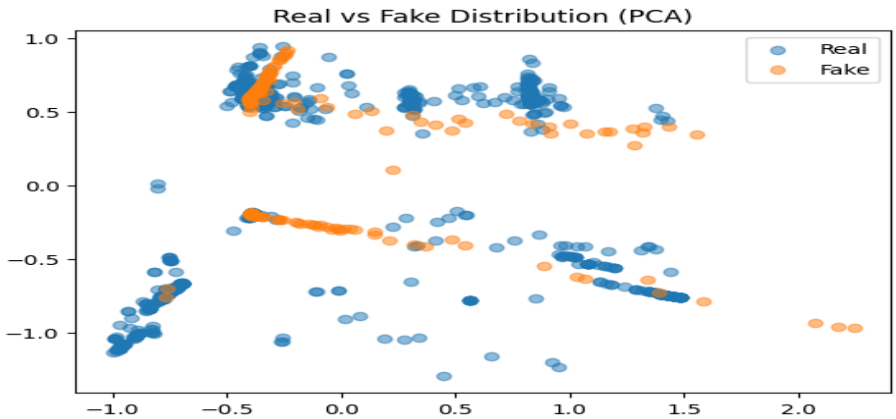

Figure 8: Graph of principal component analysis (PCA) showing real versus created network traffic

Dimensionality reduction methods, such as Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE), were utilized to evaluate the quality of the generated data by displaying the distribution of authentic and manufactured samples. **Figure 8** illustrates that the PCA projection indicates a partial overlap between the generated data and the original data distribution, implying that the GAN has absorbed specific underlying patterns. However, discrepancies in density and dispersion suggest that the model fails to accurately depict the data structure. Similarly, the t-SNE visualization in **Figure 9** provides a more intricate representation of local interactions across samples. While some grouping patterns are preserved, many gaps and inconsistencies remain between actual and generated data points.

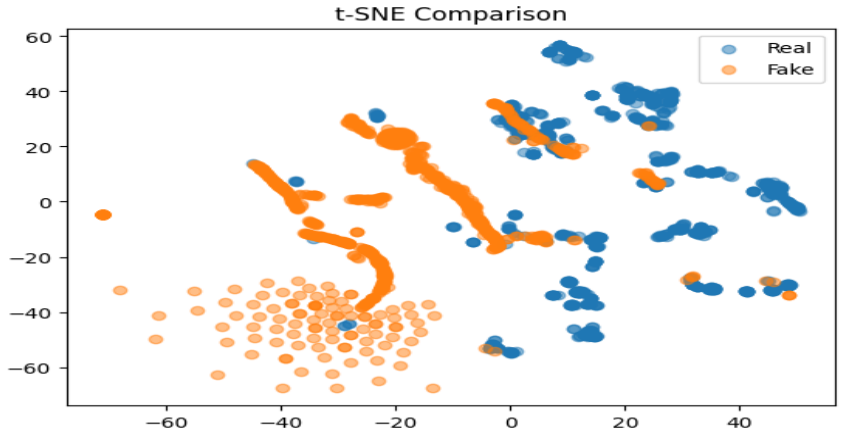

Figure 9: t-SNE visualisation of authentic versus synthetic network traffic

This experiment demonstrates that employing GANs for tabular cybersecurity data poses more significant hurdles than for image-based tasks. The increased complexity and lack of spatial organization impede the generator's capacity to accurately represent the data distribution. Although the GAN can identify some general patterns, it struggles to reproduce the complex details of the data. Improvements, such as advanced designs, enhanced feature preprocessing, or various GAN iterations, may provide higher performance.

6. Creative AI – QuickDraw Pizza GAN:

We used the QuickDraw dataset to teach a Generative Adversarial Network how to make fake pizza pictures. This dataset is different from the previous ones because it contains simple hand-drawn pictures of things. We chose the "pizza" category because its round shape and different toppings and painting techniques made it a good way to test the model's ability to learn both structure and diversity.

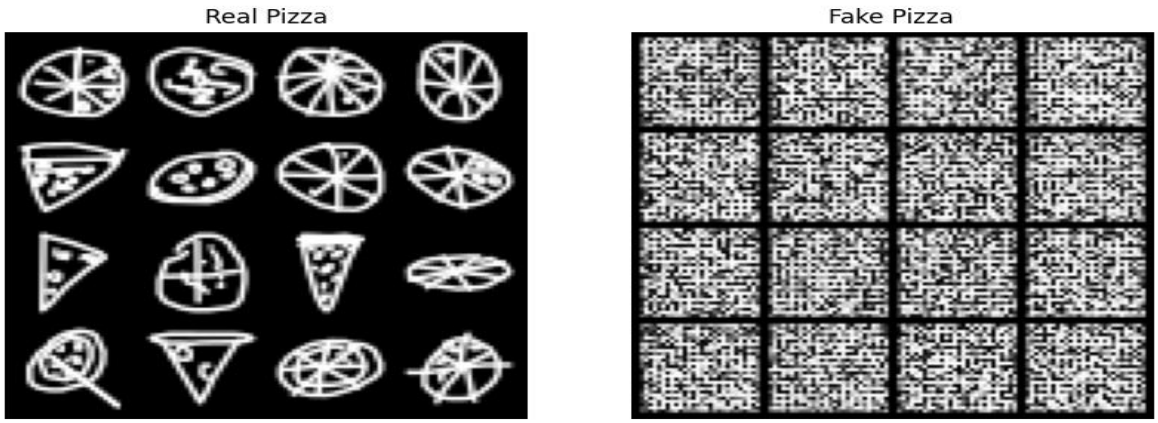

Figure 10: Comparison of real and generated pizza sketches

Figure 10 shows a side-by-side comparison of real sketches from the dataset with pictures made by the GAN. The model successfully learned the basic round shape that is common in pizza pictures. A lot of the photographs that were made look like pizzas, which means that the generator was able to capture the basic features of the dataset. But the intricate features, such toppings and designs on the inside, are not always clear and can look blurry or distorted. This means that the model understood the basic framework, but it had trouble copying the more complex variations that were in the original sketches.

We utilised a Deep Convolutional GAN (DCGAN) architecture for this job since convolutional layers are good at getting spatial characteristics from picture data. The generator was designed to convert random noise vectors into sketch-like outputs, whereas the discriminator's objective was to differentiate between authentic and generated sketches. We trained both networks at the same time using binary cross-entropy loss and the Adam optimiser.

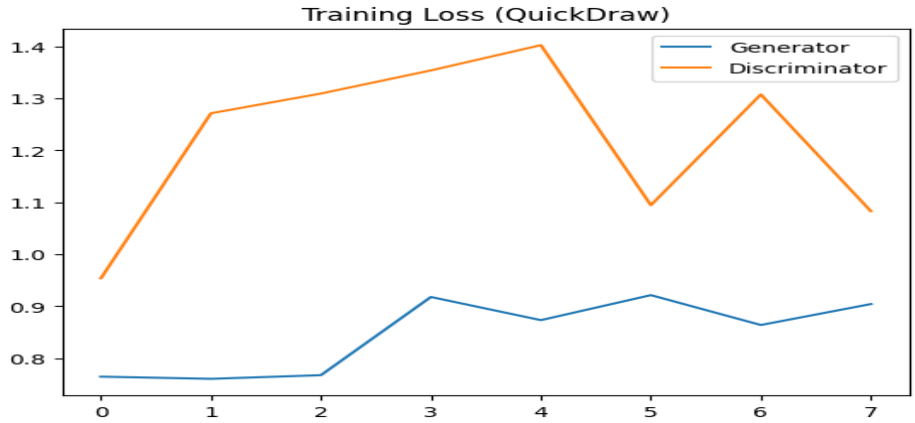

Figure 11: Training loss curves for generator and discriminator (QuickDraw dataset)

Figure 11 shows that the training loss curves change over time, which is a normal part of GAN models because they are adversarial. The generator and discriminator are always trying to outdo each other, which makes convergence unstable. Still, the general pattern shows that the model was able to learn important things from the dataset over time.

This experiment shows that GANs can make simple sketch-based visuals with shapes that can be recognised. But the outputs that are generated are still not very diverse or high-quality, especially when it comes to capturing little details and different sketching styles. This shows how hard it is to use GANs on datasets that have a lot of variation, even when the visual complexity seems low.

7. Discussion and Future Work:

The results indicate that GANs exhibit variable performance contingent upon the complexity of the dataset. The model acquired the fundamental pattern of simple data, such as the sine-wave distribution, and generated accurate samples. Nevertheless, when addressing intricate distributions such as the Gaussian mixture, the GAN encountered difficulties in capturing all modes, resulting in issues such as insufficient learning and mode collapse.

The GAN generated outputs that were readily observable in image-based challenges such as BloodMNIST, although it lacked numerous intricate features. This indicates that the stability of training and the model's capacity were constrained. In the cybersecurity challenge, the

model identified overarching patterns among the data. Nonetheless, PCA and t-SNE visualisations indicated that the generated samples did not align precisely with the real data distribution. This illustrates the difficulty of modelling high-dimensional tabular data.

The GAN acquired the fundamental framework of pizza sketches in the QuickDraw dataset, although it exhibited limited diversity and detail. This indicates that various drawing styles can significantly complicate matters, even with seemingly straightforward facts.

Future enhancements may involve the utilisation of more sophisticated GAN variants, such as Wasserstein GAN (WGAN) or Conditional GAN (cGAN), which enhance training stability and provide greater control over the output. Enhancing performance could be achieved by augmenting training data and refining preprocessing techniques. In conclusion, GANs possess significant promise; yet, they require meticulous calibration to achieve optimal outcomes.

8. Conclusion:

This study examined the utilisation of Generative Adversarial Networks in several fields, such as synthetic data modelling, medical imaging, cybersecurity, and creative sketch production. The findings indicate that GANs are proficient in learning fundamental data distributions and producing realistic examples, especially for structured and image-centric datasets.

The trials demonstrated that GANs excel with simple and continuous data, exemplified by the sine-wave distribution, where the model effectively captured the overarching pattern. Nonetheless, performance diminished with heightened data complexity, as evidenced in the Gaussian mixture and cybersecurity datasets, where the model encountered difficulties in adequately representing multi-modal and high-dimensional distributions. In image-based applications, namely utilising the BloodMNIST dataset, convolutional architectures facilitated the creation of visually significant samples; nonetheless, issues such as blurriness and insufficient fine resolution persisted.

In the creative QuickDraw job, the GAN successfully learned fundamental structural aspects, such as the round form of pizza designs, although encountered difficulties in representing diversity and complex details. These findings underscore that although GANs are potent generative models, their efficacy is significantly influenced by the characteristics of the data, the selected architecture, and the stability of the training regimen.

This work underscores the considerable potential of GANs for practical applications, including data augmentation, simulation, and creative content development. Nonetheless, obstacles like as mode collapse, training instability, and restricted output diversity persist as significant limits. Enhancing designs, training methodologies, and assessment procedures will be crucial for the progression of GANs in intricate domains.

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